TabuLLM: AI Benchmark Report

1. Executive Summary: Consolidated Audit Results

This report consolidates the results of the TabuLLM data augmentation pipeline. The objective is to audit the fidelity, diversity, and privacy of the synthetic data generated via LLM inference. Below are the averaged performance indicators across all processed datasets in this session.

Audit Indicator	Global Session Mean (± Std)
Synthetic Fidelity (JSD)	88.6 ± 1.20
Correlation Stability	92.3 ± 0.46
Distribution Consistency	90.6 ± 1.11
Entropy Preservation	88.7 ± 0.46
Mean Z Score Drift	0.1 ± 0.02
Categorical Fidelity	0.0 ± 0.00
Privacy Risk	0.0 ± 0.00
Data Density	100.0 ± 0.00
Coverage Score	98.7 ± 0.78
Feature Accuracy	95.0 ± 1.26
Novelty Score	0.8 ± 0.01

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Dataset Audit: heart_disease.csv

1. Dataset Portrait

Attribute	Value
Rows	303
Columns	14
Numeric features	14
Categorical features	0
Missing cells (%)	0.1%

2. Generation Strategy & LLM Config [1]

Parameter	Configuration
Algorithm (*12)	gpt-4o-mini
Folds (*13)	10
Context % (*14)	0%
Generation % (*15)	100%
Join context (*16)	False

3. Statistical Significance & Data Drift

Feature	Orig Mean	Aug Mean	Shift	P-Value	Status
age	54.439	55.066	+0.627	1.2%	OK
sex	0.680	0.674	-0.006	0.9%	OK
cp	3.158	3.236	+0.077	2.4%	OK
trestbps	131.690	131.331	-0.359	0.3%	OK
chol	246.693	249.430	+2.737	1.1%	OK
fbs	0.149	0.207	+0.058	39.1%	Drift
restecg	0.990	1.012	+0.022	2.3%	OK
thalach	149.607	149.533	-0.074	0.0%	OK
exang	0.327	0.343	+0.016	5.0%	OK
oldpeak	1.040	1.055	+0.016	1.5%	OK
slope	1.601	1.636	+0.036	2.2%	OK
ca	0.672	0.678	+0.005	0.8%	OK

4. Quality & Fidelity Metrics (Mean ± Std)

Metric Category	Aggregated Score
Synthetic Fidelity (JSD) (*1)	88.6 ± 1.20
Correlation Stability (*2)	92.3 ± 0.46
Distribution Consistency (*3)	90.6 ± 1.11
Entropy Preservation (*4)	88.7 ± 0.46
Mean Z Score Drift (*5)	0.1 ± 0.02
Categorical Fidelity (*6)	0.0 ± 0.00
Privacy Risk (*7)	0.0 ± 0.00

Data Density (*8)	100.0 ± 0.00
Coverage Score (*9)	98.7 ± 0.78
Feature Accuracy (*10)	95.0 ± 1.26
Novelty Score (*11)	0.8 ± 0.01

*[1] (*12) Algorithm: Contextual constraint or model parameter.*

*[2] (*13) Folds: Contextual constraint or model parameter.*

*[3] (*14) Context %: Contextual constraint or model parameter.*

*[4] (*15) Generation %: Contextual constraint or model parameter.*

*[5] (*16) Join context: Contextual constraint or model parameter.*

[6] Synthetic Fidelity (JSD): Overall distribution similarity.
[7] Correlation Stability: Multivariate relationships preserved.
[8] Distribution Consistency: Median feature similarity.
[9] Entropy Preservation: Information diversity maintained.
[10] Mean Z Score Drift: Standardized distribution shift.
[11] Categorical Fidelity: Category distribution similarity.
[12] Privacy Risk: Synthetic rows too close to originals.
[13] Data Density: Non-missing cells in augmented data.
[14] Coverage Score: Synthetic coverage of original space.
[15] Feature Accuracy: Mean feature similarity.
[16] Novelty Score: Distance ratio to nearest neighbors.

6. Downstream ML Utility: Comparative Analysis

Model: GB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.5426229508196722	0.5508196721311475	0.5737704918032788	0.03114754098360653	
F1	0.5306106235096497	0.5016085427979499	0.5517302204364427	0.02111959692679317	

Model: XGB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.5426229508196723	0.5590163934426229	0.5557377049180328	0.01311475409836064	
F1	0.5198013329305267	0.5209322712126329	0.5313748572439251	0.01157352431339830 9	

Model: DT

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.5213114754098364	0.5278688524590164	0.4950819672131148	-0.026229508196721308	
F1	0.5254388871184479	0.512971199393186	0.4984397590493294 7	-0.026999128069118528	

Model: NB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.5573770491803279	0.5573770491803278	0.5704918032786884	0.01311475409836067	
F1	0.568660979990255	0.5540097892985251	0.5690883489007403	0.00042736891048530403	

Model: SVM

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.5475409836065573	0.5475409836065573	0.5475409836065573	0	
F1	0.3899270600110827	0.3899270600110827	0.3899270600110827	0	

Model: AB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6081967213114754	0.560655737704918	0.5950819672131147	-0.01311475409836066	
F1	0.6070236204175147	0.526684774616352 9	0.566421077860554	-0.04060254255696064	

Model: RF

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.5885245901639344	0.5704918032786885	0.601639344262295	0.01311475409836066	
F1	0.5537742023485703	0.4935728001142004	0.5463036871620772	-0.00747051518649326 5	

Model: LR

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.5983606557377049	0.5918032786885246	0.601639344262295	0.00327868852459014 53	
F1	0.5648347697855012	0.5266829068124631	0.5519142622101931	-0.01292050757530802 6	

Model: MLP

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.5311475409836064	0.5557377049180328	0.5918032786885247	0.06065573770491805 6	
F1	0.4674008535102789 4	0.504687017112778	0.5287083921585354	0.06130753864825637 4	

Model: KNN

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.5540983606557377	0.5426229508196722	0.5426229508196722	-0.01147540983606557	
F1	0.4442202071436568 4	0.4241227845144789 3	0.4431891347173654	-0.00103107242629150 17	

Appendix: Benchmark Data Trace

Legend: [Blue] = Real Context | [Green] = Generated Synthetic | [Yellow] = Real Training (Remaining) | [Grey] = Real Test Baseline

A. Baseline: Original Dataset (Full Record Sample)

#	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak
1	63.0	1.0	1.0	145.0	233.0	1.0	2.0	150.0	0.0	2.3
2	67.0	1.0	4.0	160.0	286.0	0.0	2.0	108.0	1.0	1.5
3	67.0	1.0	4.0	120.0	229.0	0.0	2.0	129.0	1.0	2.6
4	37.0	1.0	3.0	130.0	250.0	0.0	0.0	187.0	0.0	3.5
5	41.0	0.0	2.0	130.0	204.0	0.0	2.0	172.0	0.0	1.4
6	56.0	1.0	2.0	120.0	236.0	0.0	0.0	178.0	0.0	0.8
7	62.0	0.0	4.0	140.0	268.0	0.0	2.0	160.0	0.0	3.6
8	57.0	0.0	4.0	120.0	354.0	0.0	0.0	163.0	1.0	0.6
9	63.0	1.0	4.0	130.0	254.0	0.0	2.0	147.0	0.0	1.4
10	53.0	1.0	4.0	140.0	203.0	1.0	2.0	155.0	1.0	3.1
11	57.0	1.0	4.0	140.0	192.0	0.0	0.0	148.0	0.0	0.4
12	56.0	0.0	2.0	140.0	294.0	0.0	2.0	153.0	0.0	1.3
13	56.0	1.0	3.0	130.0	256.0	1.0	2.0	142.0	1.0	0.6
14	44.0	1.0	2.0	120.0	263.0	0.0	0.0	173.0	0.0	0.0
15	52.0	1.0	3.0	172.0	199.0	1.0	0.0	162.0	0.0	0.5
16	57.0	1.0	3.0	150.0	168.0	0.0	0.0	174.0	0.0	1.6
17	48.0	1.0	2.0	110.0	229.0	0.0	0.0	168.0	0.0	1.0
18	54.0	1.0	4.0	140.0	239.0	0.0	0.0	160.0	0.0	1.2
19	48.0	0.0	3.0	130.0	275.0	0.0	0.0	139.0	0.0	0.2
20	49.0	1.0	2.0	130.0	266.0	0.0	0.0	171.0	0.0	0.6

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B.1 Audit Trace: Fold 1 | Gen: 100%

#	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak
1	60.0	1.0	2.0	129.0	256.0	0.0	2.0	159.0	0.0	0.5724682200908678
2	50.0	1.0	4.0	168.0	284.0	1.0	2.0	139.0	0.0	-1.1358939342053687
3	61.0	1.0	4.0	137.0	138.0	1.0	0.0	135.0	1.0	0.5313305713351711
4	48.0	1.0	4.0	111.0	246.0	0.0	2.0	152.0	0.0	0.5189423144831465
5	67.0	1.0	1.0	179.0	316.0	0.0	2.0	134.0	0.0	0.6113219683395912
6	68.0	0.0	4.0	127.0	268.0	1.0	2.0	162.0	0.0	0.0
7	49.0	0.0	4.0	130.0	269.0	0.0	0.0	163.0	0.0	0.0
8	54.0	1.0	3.0	120.0	258.0	0.0	2.0	147.0	0.0	0.4
9	55.0	1.0	2.0	130.0	262.0	0.0	0.0	155.0	0.0	0.0
10	46.0	1.0	2.0	101.0	197.0	1.0	0.0	156.0	0.0	0.0

B.2 Audit Trace: Fold 2 | Gen: 100%

#	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak
1	61.0	0.0	3.0	110.0	265.0	0.0	0.0	176.0	0.0	0.5796986441697325
2	55.0	1.0	3.0	136.0	285.0	0.0	2.0	128.0	0.0	0.06549865936776908
3	53.0	0.0	4.0	149.0	314.0	1.0	0.0	165.0	0.0	0.2692507644932624
4	54.0	1.0	4.0	135.0	170.0	0.0	2.0	133.0	1.0	2.950408693428641 5
5	45.0	1.0	1.0	127.0	224.0	0.0	0.0	127.0	1.0	2.8635106978259506
6	61.0	0.0	4.0	139.0	305.0	0.0	2.0	139.0	0.0	4.420985506864497
7	63.0	0.0	3.0	135.0	252.0	0.0	2.0	172.0	0.0	0.0
8	60.0	1.0	4.0	130.0	253.0	0.0	0.0	144.0	1.0	1.4
9	58.0	0.0	2.0	136.0	319.0	1.0	2.0	152.0	0.0	0.0
10	59.0	0.0	4.0	174.0	249.0	0.0	0.0	143.0	1.0	0.0

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B.3 Audit Trace: Fold 3 | Gen: 100%

#	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak
1	64.0	0.0	4.0	152.0	197.0	0.0	0.0	106.0	1.0	3.2803285167210525
2	48.0	1.0	2.0	119.0	303.0	0.0	0.0	161.0	0.0	-0.021746396603389426
3	60.0	1.0	4.0	129.0	251.0	0.0	0.0	138.0	0.0	1.474801275118725
4	51.0	1.0	1.0	125.0	280.0	0.0	2.0	134.0	0.0	-0.06761109316141323
5	53.0	1.0	1.0	148.0	224.0	1.0	2.0	162.0	0.0	-0.23511677415963217
6	52.0	1.0	2.0	130.0	265.0	0.0	2.0	140.0	0.0	1.552041321139425
7	49.0	1.0	3.0	120.0	188.0	0.0	0.0	139.0	0.0	2.0
8	45.0	1.0	4.0	115.0	260.0	0.0	2.0	185.0	0.0	0.0
9	45.0	0.0	2.0	130.0	234.0	0.0	2.0	175.0	0.0	0.6
10	46.0	1.0	3.0	150.0	231.0	0.0	0.0	147.0	0.0	3.6

B.4 Audit Trace: Fold 4 | Gen: 100%

#	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak
1	64.0	1.0	4.0	115.0	265.0	0.0	2.0	127.0	0.0	0.652353377386079
2	40.0	1.0	3.0	117.0	225.0	0.0	0.0	197.0	0.0	-0.3305725575500644
3	64.0	0.0	4.0	101.0	242.0	0.0	0.0	111.0	0.0	1.4295944699943175
4	49.0	1.0	4.0	103.0	336.0	0.0	0.0	155.0	1.0	0.33333239756505095
5	66.0	1.0	4.0	138.0	193.0	0.0	0.0	124.0	1.0	0.6421620477304664
6	54.0	0.0	1.0	86.0	244.0	0.0	2.0	126.0	0.0	5.052620644064849
7	57.0	1.0	4.0	140.0	192.0	0.0	0.0	148.0	0.0	0.4
8	63.0	0.0	4.0	150.0	407.0	0.0	2.0	154.0	0.0	4.0
9	57.0	0.0	4.0	128.0	303.0	0.0	2.0	159.0	0.0	0.0
10	60.0	1.0	4.0	117.0	230.0	1.0	0.0	160.0	1.0	1.4

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B.5 Audit Trace: Fold 5 | Gen: 100%

#	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak
1	44.0	1.0	3.0	125.0	217.0	0.0	2.0	120.0	1.0	2.551381485164712
2	62.0	1.0	4.0	113.0	220.0	0.0	2.0	135.0	0.0	2.1352859991072357
3	50.0	1.0	3.0	134.0	194.0	0.0	0.0	146.0	1.0	-0.04044708795468044
4	60.0	0.0	4.0	162.0	289.0	0.0	2.0	142.0	0.0	3.777536853119155
5	48.0	1.0	4.0	124.0	248.0	0.0	2.0	164.0	1.0	3.1381794109906034
6	46.0	0.0	1.0	104.0	225.0	0.0	0.0	132.0	0.0	0.23127668477347063
7	50.0	0.0	2.0	120.0	244.0	0.0	0.0	162.0	0.0	1.1
8	56.0	1.0	2.0	120.0	236.0	0.0	0.0	178.0	0.0	0.8
9	61.0	1.0	3.0	150.0	243.0	1.0	0.0	137.0	1.0	1.0
10	64.0	0.0	3.0	140.0	313.0	0.0	0.0	133.0	0.0	0.2

B.6 Audit Trace: Fold 6 | Gen: 100%

#	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak
1	49.0	0.0	2.0	125.0	271.0	0.0	0.0	160.0	1.0	0.8516042473065674
2	44.0	1.0	1.0	122.0	239.0	0.0	0.0	154.0	0.0	0.049849752102712475
3	74.0	0.0	3.0	131.0	323.0	0.0	0.0	171.0	0.0	-0.669162210071905
4	42.0	0.0	4.0	105.0	308.0	0.0	2.0	161.0	0.0	0.25197067016628
5	59.0	1.0	3.0	152.0	280.0	1.0	2.0	152.0	0.0	0.121064867565193
6	35.0	1.0	4.0	149.0	227.0	0.0	2.0	159.0	0.0	4.029077220553954
7	39.0	1.0	4.0	118.0	219.0	0.0	0.0	140.0	0.0	1.2
8	62.0	0.0	4.0	138.0	294.0	1.0	0.0	106.0	0.0	1.9
9	66.0	1.0	4.0	160.0	228.0	0.0	2.0	138.0	0.0	2.3
10	39.0	1.0	3.0	140.0	321.0	0.0	2.0	182.0	0.0	0.0

B.7 Audit Trace: Fold 7 | Gen: 100%

#	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak
1	42.0	1.0	2.0	113.0	246.0	0.0	0.0	186.0	0.0	0.0
2	60.0	1.0	2.0	142.0	241.0	0.0	0.0	172.0	0.0	0.040262926111637 11
3	58.0	0.0	4.0	111.0	201.0	0.0	2.0	151.0	0.0	0.061687830823866 59
4	51.0	1.0	4.0	129.0	281.0	0.0	0.0	154.0	1.0	0.622692456988884 4
5	51.0	1.0	4.0	129.0	213.0	0.0	0.0	173.0	0.0	4.063923909638511
6	63.0	1.0	4.0	151.0	242.0	1.0	2.0	130.0	1.0	-0.030796016011130 467
7	56.0	0.0	4.0	134.0	409.0	0.0	2.0	150.0	1.0	1.9
8	58.0	1.0	3.0	105.0	240.0	0.0	2.0	154.0	1.0	0.6
9	54.0	1.0	2.0	192.0	283.0	0.0	2.0	195.0	0.0	0.0
10	57.0	1.0	4.0	110.0	201.0	0.0	0.0	126.0	1.0	1.5

B.8 Audit Trace: Fold 8 | Gen: 100%

#	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak
1	59.0	1.0	3.0	118.0	253.0	0.0	2.0	144.0	0.0	-0.016118775103259 098
2	48.0	1.0	3.0	144.0	231.0	0.0	2.0	165.0	0.0	1.312322117429630 2
3	45.0	1.0	3.0	135.0	242.0	0.0	0.0	121.0	0.0	3.377084872303452 6
4	57.0	1.0	4.0	126.0	191.0	0.0	0.0	167.0	0.0	0.0
5	64.0	0.0	4.0	151.0	234.0	0.0	0.0	165.0	1.0	0.809445626761564 2
6	48.0	0.0	4.0	167.0	176.0	1.0	2.0	138.0	1.0	-0.080029307386829 3
7	52.0	1.0	1.0	118.0	186.0	0.0	2.0	190.0	0.0	0.0
8	54.0	0.0	3.0	135.0	304.0	1.0	0.0	170.0	0.0	0.0
9	61.0	1.0	4.0	140.0	207.0	0.0	2.0	138.0	1.0	1.9
10	61.0	1.0	3.0	150.0	243.0	1.0	0.0	137.0	1.0	1.0

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B.9 Audit Trace: Fold 9 | Gen: 100%

#	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak
1	61.0	1.0	4.0	138.0	259.0	0.0	0.0	119.0	1.0	3.944615377383979
2	53.0	1.0	4.0	137.0	201.0	0.0	0.0	163.0	0.0	1.5254516815040908
3	69.0	1.0	4.0	150.0	219.0	0.0	0.0	160.0	1.0	1.4393772176275705
4	70.0	1.0	1.0	166.0	288.0	0.0	2.0	153.0	0.0	1.2883664752370112
5	59.0	1.0	4.0	135.0	192.0	0.0	0.0	110.0	1.0	3.1743835189686832
6	62.0	1.0	3.0	127.0	230.0	0.0	0.0	164.0	0.0	1.7951256204922117
7	44.0	0.0	3.0	118.0	242.0	0.0	0.0	149.0	0.0	0.3
8	38.0	1.0	1.0	120.0	231.0	0.0	0.0	182.0	1.0	3.8
9	49.0	1.0	3.0	120.0	188.0	0.0	0.0	139.0	0.0	2.0
10	61.0	1.0	4.0	120.0	260.0	0.0	0.0	140.0	1.0	3.6

B.10 Audit Trace: Fold 10 | Gen: 100%

#	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak
1	68.0	1.0	4.0	123.0	306.0	0.0	0.0	172.0	0.0	0.3221191091537149
2	41.0	1.0	3.0	149.0	281.0	0.0	2.0	182.0	0.0	0.07065295585597803
3	69.0	1.0	4.0	106.0	237.0	0.0	2.0	117.0	0.0	1.1957125257629797
4	60.0	1.0	3.0	108.0	283.0	0.0	0.0	180.0	1.0	1.6908099999416204
5	79.0	1.0	4.0	141.0	368.0	0.0	0.0	123.0	0.0	2.304067942276398
6	62.0	1.0	1.0	135.0	285.0	0.0	0.0	162.0	0.0	0.3269039460564384 7
7	63.0	0.0	3.0	135.0	252.0	0.0	2.0	172.0	0.0	0.0
8	63.0	1.0	1.0	145.0	233.0	1.0	2.0	150.0	0.0	2.3
9	60.0	1.0	4.0	125.0	258.0	0.0	2.0	141.0	1.0	2.8
10	62.0	0.0	4.0	140.0	268.0	0.0	2.0	160.0	0.0	3.6